Signature-Based Deep Hedging for Path-Dependent Options

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Abstract

The traditional framework for derivative hedging, which relies on the local sensitivities of the Black-Scholes model, becomes less reliable in the presence of market frictions such as transaction costs and discrete trading. Deep Hedging has emerged as a flexible alternative because it formulates hedging as a data-driven stochastic control problem rather than as exact replication [3]. However, for path-dependent derivatives such as Asian options, feeding raw price histories into neural networks can create high-dimensional inputs and increase the risk of overfitting.

This paper studies whether rough path signatures can provide a compact and informative representation of price histories for deep hedging. The signature transform maps an observed path into a fixed-dimensional vector of iterated integrals, preserving ordered path information while reducing the dependence on raw sequential inputs [4, 2]. I apply this idea to the hedging of a discretely monitored Asian call option under proportional transaction costs.

Using simulated Geometric Brownian Motion paths, I compare a standard feed-forward neural network using raw price history features against a signature-based feed-forward network using truncated path signatures. In the numerical experiment, the signature-based model reduces the out-of-sample 99% Conditional Value-at-Risk (CVaR) of terminal shortfall by approximately 27.4% relative to the raw-path benchmark, while using fewer trainable parameters. These results suggest that signatures can improve the efficiency and robustness of deep hedging for path-dependent derivatives.

1 Introduction and Problem Formulation

1.1 The Discrete Trading Framework

The classical Black-Scholes framework relies on the assumption of a frictionless, continuous-time market where perfect replication of contingent claims is theoretically possible. In practice, trading occurs in discrete time and incurs transaction costs. To formalize a more realistic hedging environment, we use a discrete trading framework.

Let $T > 0$ be a finite time horizon and let

$$\mathcal{T} = \{t_0, t_1, \ldots, t_N = T\},$$

where $t_0 = 0$. The market consists of a risk-free asset, normalized to one under a zero-interest-rate assumption, and a risky asset with price process

$$S = (S_{t_k})_{k=0}^{N}.$$

The process is adapted to a discrete-time filtration

$$\mathbb{F} = (\mathcal{F}_{t_k})_{k=0}^{N},$$

where $\mathcal{F}_{t_k}$ represents the information available up to time t_k.

A trading strategy is a predictable process

$$\delta = (\delta_{t_k})_{k=0}^{N-1},$$

where δ_{t_k} denotes the number of shares held over $(t_k, t_{k+1}]$. Predictability means that δ_{t_k} is $\mathcal{F}_{t_k}$-measurable, so the hedge at time t_k only uses information available at that time.

In a frictionless market, the terminal wealth of a self-financing portfolio initialized with capital V_0 is

$$V_T(\delta) = V_0 + \sum_{k=0}^{N-1} \delta_{t_k} \left(S_{t_{k+1}} - S_{t_k} \right). \tag{1}$$

1.2 Introducing Market Frictions

Continuous delta hedging becomes problematic under proportional transaction costs. Since Brownian paths have infinite variation, continuously rebalancing a Black-Scholes delta hedge can generate unbounded trading costs. This motivates a discrete-time, cost-aware formulation.

Let $c \in (0, 1)$ be a proportional transaction cost rate. Every change in position incurs a cost proportional to the traded notional. We set the initial position to $\delta_{t_{-1}} = 0$ and impose liquidation at maturity by defining $\delta_{t_N} = 0$. The terminal wealth with transaction costs is

$$V_T^c(\delta) = V_0 + \sum_{k=0}^{N-1} \delta_{t_k} \left(S_{t_{k+1}} - S_{t_k} \right) - \sum_{k=0}^{N} c S_{t_k} \left| \delta_{t_k} - \delta_{t_{k-1}} \right|. \tag{2}$$

The cost term breaks exact replication. Therefore, the objective changes from perfect replication to risk minimization.

1.3 The Deep Hedging Formulation

Let Z be the non-negative, $\mathcal{F}_T$-measurable payoff of a derivative contract. The terminal hedging shortfall is defined as

$$L_T(\delta) = Z - V_T^c(\delta). \tag{3}$$

Positive values of L_T represent losses relative to the derivative liability.

The goal is to find a predictable strategy δ^* that minimizes a risk measure of the terminal shortfall:

$$\delta^* \in \arg\min_{\delta} \rho(L_T(\delta)). \tag{4}$$

Following the Deep Hedging framework of Buehler et al. [3], I use Conditional Value-at-Risk (CVaR), also known as Expected Shortfall, as the risk criterion. For confidence level $\alpha \in (0, 1)$,

$$\mathrm{CVaR}_\alpha(L_T(\delta)) = \inf_{q \in \mathbb{R}} \left\{ q + \frac{1}{1-\alpha} \mathbb{E}\left[(L_T(\delta) - q)^+ \right] \right\}. \tag{5}$$

Deep Hedging parameterizes the strategy with neural networks. At each trading date t_k,

$$\delta_{t_k} = F^{\theta_k}(I_{t_k}), \tag{6}$$

where I_{t_k} is the feature vector available at time t_k and θ_k are trainable parameters. The full parameter vector $\theta = (\theta_0, \ldots, \theta_{N-1})$ is optimized by stochastic gradient descent.

1.4 The Path-Dependency Bottleneck

For a European call option, $Z = (S_T - K)^+$ depends only on the terminal price. In a Markovian setting, the hedge can be based on the current spot and previous holdings:

$$I_{t_k} = (S_{t_k}, \delta_{t_{k-1}}).$$

For path-dependent options, this is no longer sufficient.

Consider a discretely monitored arithmetic Asian call option:

$$Z_{\text{Asian}} = \left(\frac{1}{N} \sum_{j=1}^{N} S_{t_j} - K \right)^+. \tag{7}$$

Because the payoff depends on the entire price path, the hedge should condition on historical information:

$$I_{t_k} = (S_{t_0}, S_{t_1}, \ldots, S_{t_k}, \delta_{t_{k-1}}). \tag{8}$$

A raw feed-forward network faces three practical issues in this setting. First, the input dimension grows with the number of monitoring dates. Second, fixed-width feed-forward networks require padding or truncation to handle different observation times. Third, high-dimensional raw paths can make the model more vulnerable to overfitting simulated noise.

This motivates a feature map that summarizes the observed path in a fixed-dimensional representation while preserving its ordered structure. Rough path signatures provide one such representation.

2 Rough Path Signatures as a Feature Extractor

2.1 Continuous Path Embedding and Definition

To apply the signature method, the discrete price observations are embedded into a continuous path. Let

$$X : [0, T] \to \mathbb{R}^d$$

be a continuous path constructed by interpolating observed state variables. In this experiment, the relevant path is time-augmented:

$$X_t = (t, S_t).$$

Including time prevents the representation from depending only on price increments and allows the model to distinguish when movements occur.

The signature of X over $[0, T]$, denoted $\mathbb{S}(X)_{0,T}$, is the infinite sequence

$$\mathbb{S}(X)_{0,T} = \left(1, \mathbf{X}^1, \mathbf{X}^2, \dots, \mathbf{X}^k, \dots\right), \tag{9}$$

where the kth level is

$$\mathbf{X}^k = \int_{0<t_1<\cdots<t_k<T} dX_{t_1} \otimes \cdots \otimes dX_{t_k}. \tag{10}$$

For a d-dimensional path, truncating the signature at depth M gives

$$\mathbb{S}^M(X)_{0,T} = \left(1, \mathbf{X}^1, \dots, \mathbf{X}^M\right). \tag{11}$$

This creates a finite-dimensional feature vector.

For intuition, the first level records total increments. The second level records ordered pairwise interactions between coordinates, including area-like terms. Higher levels capture increasingly complex ordered interactions. This makes signatures useful for machine learning on paths [4, 2].

2.2 Properties Relevant for Deep Hedging

The signature transform has two properties that are especially relevant for hedging path-dependent claims.

First, signatures preserve ordered path information. Since an Asian option depends on the full trajectory of the underlying, a useful feature representation should capture not only the current price but also how the path arrived there.

Second, signatures provide a universal feature representation for continuous functions on paths under appropriate conditions [4]. Informally, many path-dependent functionals can be approximated by functions of the signature. Thus, the hedge can be modeled as

$$\delta_{t_k} \approx f_\theta\left(\mathbb{S}^M(X)_{0,t_k}, \delta_{t_{k-1}}\right). \tag{12}$$

This replaces the growing raw input

$$(S_{t_0}, \ldots, S_{t_k})$$

with a fixed-dimensional feature vector.

Recent work has also connected signatures to stochastic control and modern sequence modeling, supporting their use as compact state representations in dynamic decision problems [1, 5]. In this paper, I use this idea in a focused deep hedging experiment rather than attempting to build a full general theory.

3 Numerical Setup and Experimental Design

To evaluate the signature representation, I simulate a discrete hedging environment where the agent minimizes the empirical 99% CVaR of terminal shortfall.

3.1 Market and Option Parameters

The underlying follows a Geometric Brownian Motion:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t. \tag{13}$$

The parameters are

$$S_0 = 100, \quad K = 100, \quad \mu = 0, \quad \sigma = 0.20, \quad T = \frac{60}{252}.$$

The option is a discretely monitored arithmetic Asian call with payoff

$$Z = \max\left(\frac{1}{60}\sum_{i=1}^{60} S_{t_i} - K, 0\right). \tag{14}$$

The model is trained on 20,000 simulated paths and evaluated out of sample on 5,000 test paths.

3.2 Hedging Constraints and Costs

The agent rebalances at each of the 60 monitoring dates. Transaction costs are proportional at rate

$$c = 0.001,$$

or 10 basis points per trade. The hedge ratio is constrained to lie in $[0, 1]$. This restriction prevents short positions and leverage, making the comparison conservative but also limiting the feasible strategy class.

The reported mean terminal PnL is defined as

$$\text{PnL} = V_T^c(\delta) - Z.$$

The risk metric is the 99% CVaR of the shortfall

$$L_T(\delta) = Z - V_T^c(\delta).$$

No explicit option premium is added to the initial wealth convention in the experiment, so the mean PnL should be interpreted as a relative hedging-performance measure rather than as a full option pricing result.

3.3 Model Architectures

Two feed-forward neural network policies are compared.

- **Raw-FFNN:** The benchmark model receives the raw price history, zero-padded to a fixed length of 60, plus the previous hedge position. Its input dimension is therefore 61.
- **Sig-FFNN:** The proposed model receives the depth-3 signature of the time-augmented path (t, S_t), computed using the `signatory` library, plus the previous hedge position. Since a two-dimensional path has 14 signature features up to depth 3, the total input dimension is 15.

Both networks use three hidden layers with 64 neurons per layer, ReLU activations, and a sigmoid output to enforce the hedge constraint. The optimizer is Adam, the batch size is 1,024, and both models are trained for 100 epochs.

4 Results and Discussion

Table 1: Out-of-Sample Hedging Performance on 5,000 GBM Test Paths

Metric	**Raw-FFNN**	**Sig-FFNN**
Epochs Run	100	100
Trainable Parameters	12,353	9,409
Mean Terminal PnL	-2.4822	-2.4904
99% CVaR of Shortfall	4.9669	3.6061

The signature-based model achieves a lower out-of-sample 99% CVaR of terminal shortfall. The Raw-FFNN obtains a CVaR of 4.9669, while the Sig-FFNN obtains 3.6061. The relative reduction is

$$\frac{4.9669 - 3.6061}{4.9669} \approx 27.4\%. \tag{15}$$

This suggests that the signature representation improves downside risk control in this simulated hedging problem.

The mean terminal PnL values are very similar: -2.4822 for the Raw-FFNN and -2.4904 for the Sig-FFNN. Therefore, the main empirical difference is not

average PnL, but tail behavior. This is consistent with the training objective, which directly targets the extreme shortfall tail.

The Sig-FFNN also uses fewer trainable parameters: 9,409 compared with 12,353 for the Raw-FFNN. This is a 23.8% reduction in model size:

$$\frac{12,353 - 9,409}{12,353} \approx 23.8\%. \tag{16}$$

The result provides evidence that the signature transform can make the learning problem more efficient by replacing a sparse, zero-padded raw history with a compact representation of ordered path information.

4.1 Performance Visualization

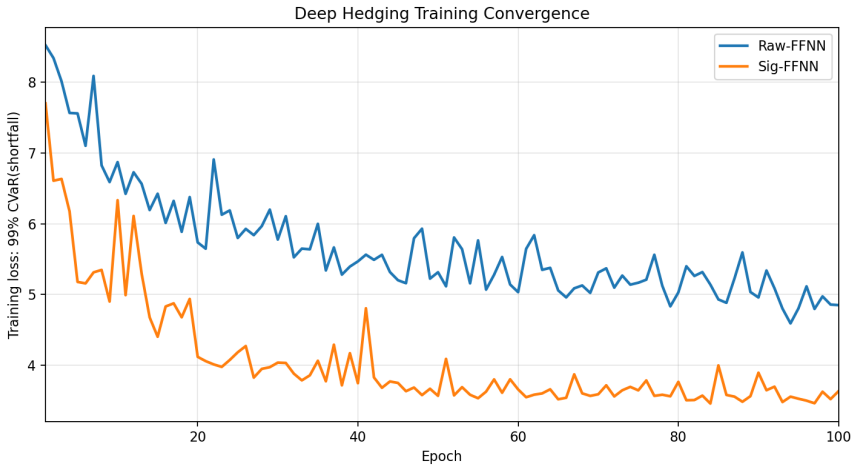


Figure 1: Training convergence: 99% CVaR of terminal shortfall on the training set. The Sig-FFNN reaches a lower final CVaR than the raw-price benchmark.

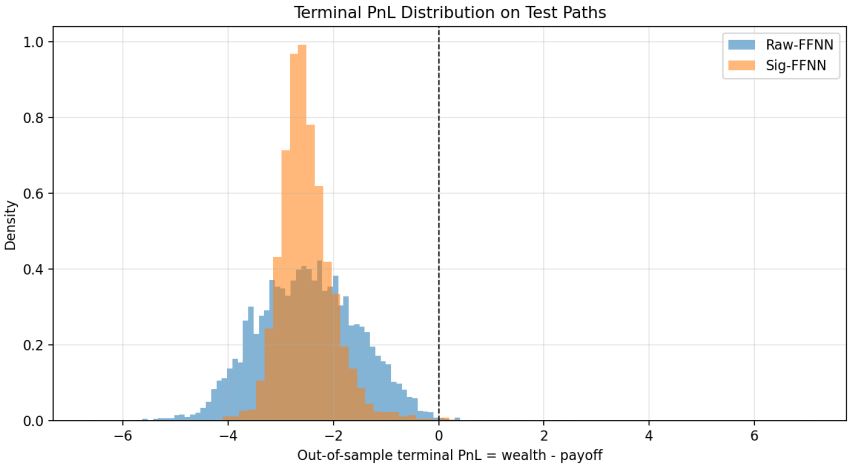


Figure 2: Out-of-sample terminal PnL distribution on 5,000 test paths. The Sig-FFNN has similar mean PnL but a better adverse-tail profile.

5 Conclusion and Future Extensions

5.1 Conclusion

This paper investigated the use of rough path signatures for deep hedging of path-dependent derivatives under proportional transaction costs. The focus was a discretely monitored Asian call option, where the payoff depends on the full price trajectory rather than only on terminal spot.

The numerical experiment compares a raw-history feed-forward neural network with a signature-based feed-forward neural network. The Sig-FFNN reduces the out-of-sample 99% CVaR of terminal shortfall from 4.9669 to 3.6061, corresponding to an approximate 27.4% reduction in tail risk. It also uses fewer trainable parameters, suggesting that the signature transform provides a compact and useful path representation.

The two models have similar mean terminal PnL, so the benefit of the signature representation appears mainly in downside risk control. This is economically meaningful because the hedging objective is not simply to improve average performance, but to reduce severe losses under transaction costs.

5.2 Limitations and Future Work

Several limitations should be emphasized. First, the experiment uses simulated GBM paths. Real financial returns may exhibit stochastic volatility, jumps, heavy tails, volatility clustering, and market microstructure effects. Second, the transaction cost model is constant and proportional, whereas real costs depend on liquidity, trade size, volatility, and bid-ask spreads. Third, the hedge ratio

is restricted to $[0, 1]$, which excludes short and leveraged positions. Fourth, the experiment uses one main market configuration. A more complete study would repeat the comparison across multiple random seeds, maturities, volatilities, strikes, transaction cost levels, and payoff types.

Future work could extend the experiment to Heston or jump-diffusion models, compare signatures against recurrent neural networks and transformers, and test whether signature features remain robust on real market data. It would also be useful to study different signature depths, since deeper signatures may contain richer path information but also increase feature dimension.

Declaration of AI Assistance

I acknowledge the use of AI coding assistants, specifically OpenAI Codex and related large language models, during the software implementation phase of this project.

The AI assistant was used to accelerate the translation of the quantitative design into PyTorch code. Specifically, it helped with structuring neural network training loops, managing tensor shapes for time-augmented paths before passing them to the `signatory` library, and debugging the empirical 99% CVaR loss function.

References

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